

Homework #7

Hyeonmin Lee

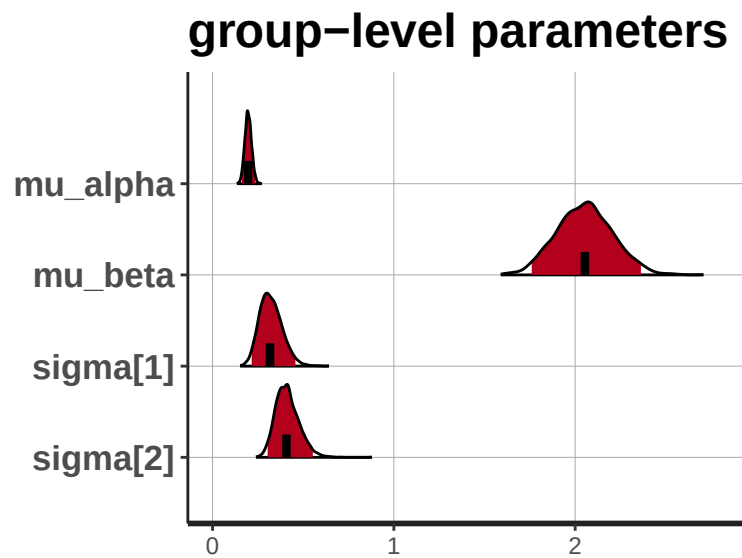
Q1.

(1)

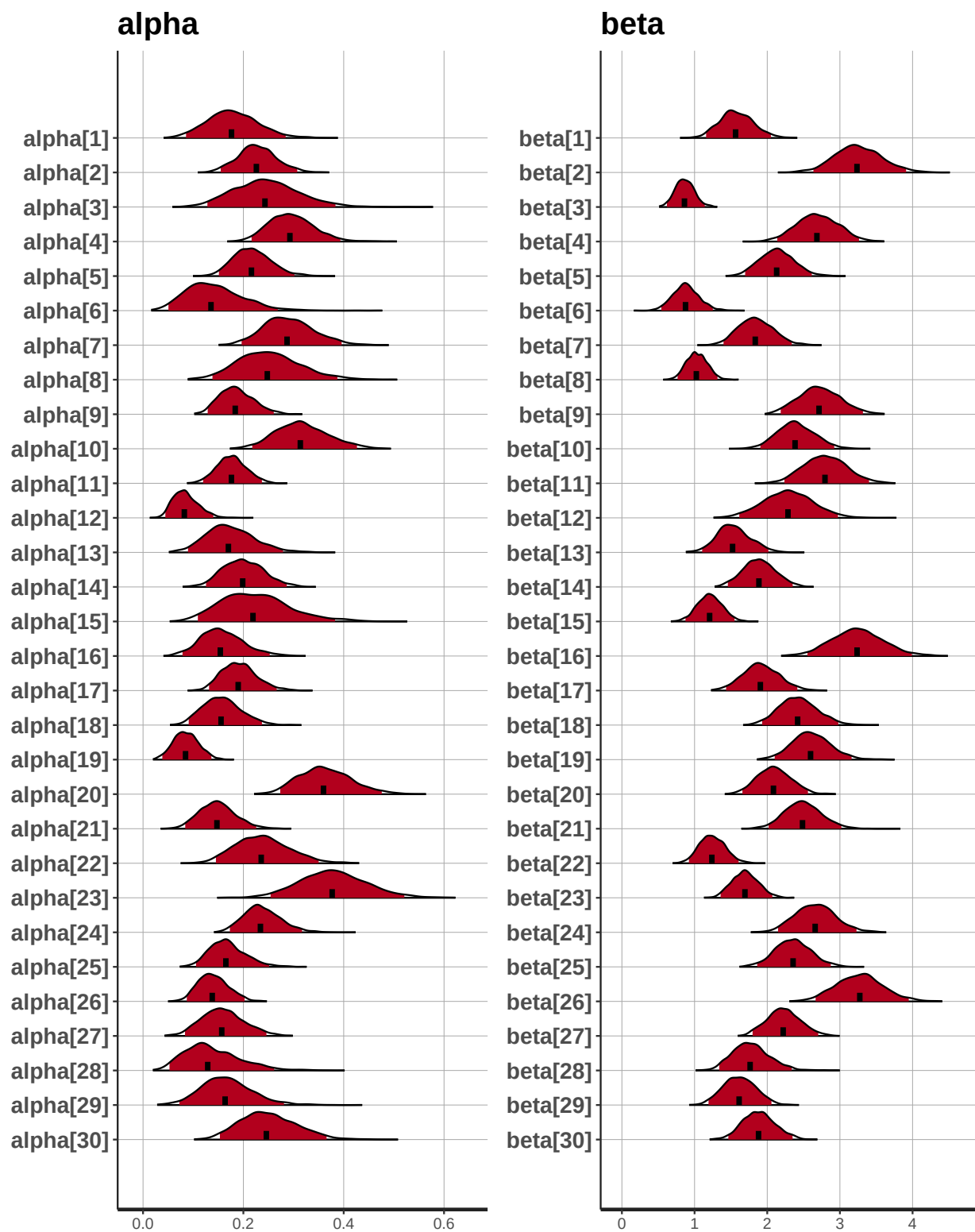
1. run `simulate_hw7_model1.R` and create `simul_data_hw7_model1.txt`
2. See codes in Q1 folder

(2a)

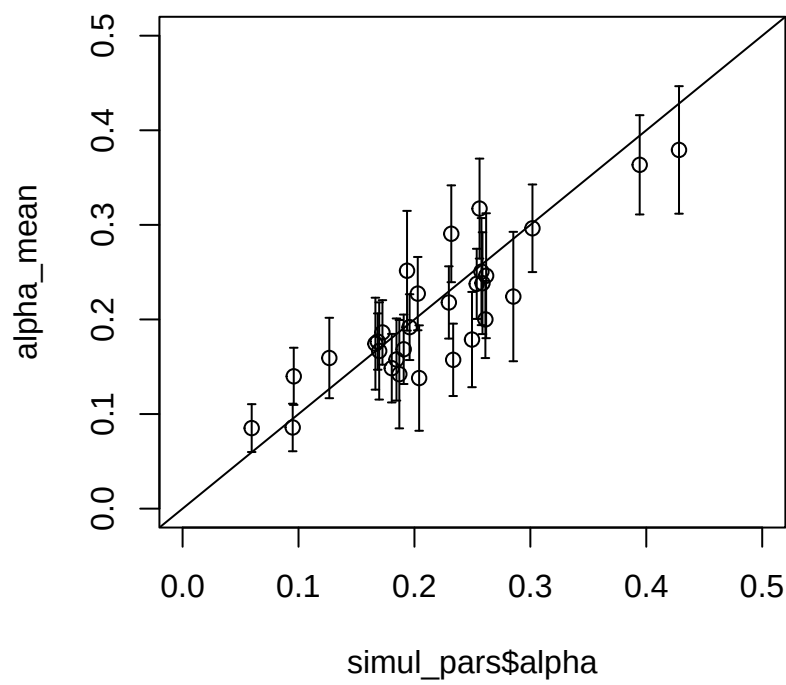
Posterior distributions of group-level parameters

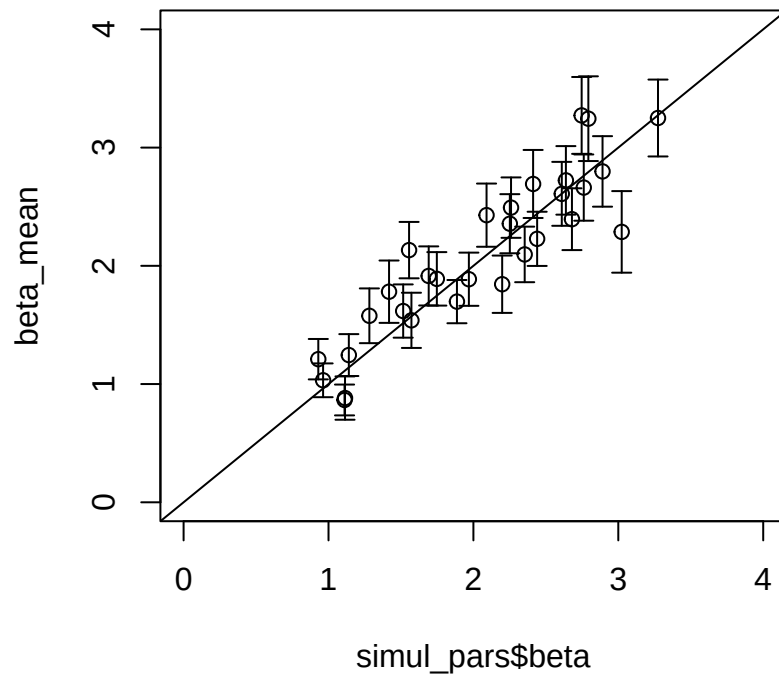


Posterior distributions of individual-level parameters



(2b) Scatterplot of true parameters and estimated parameters

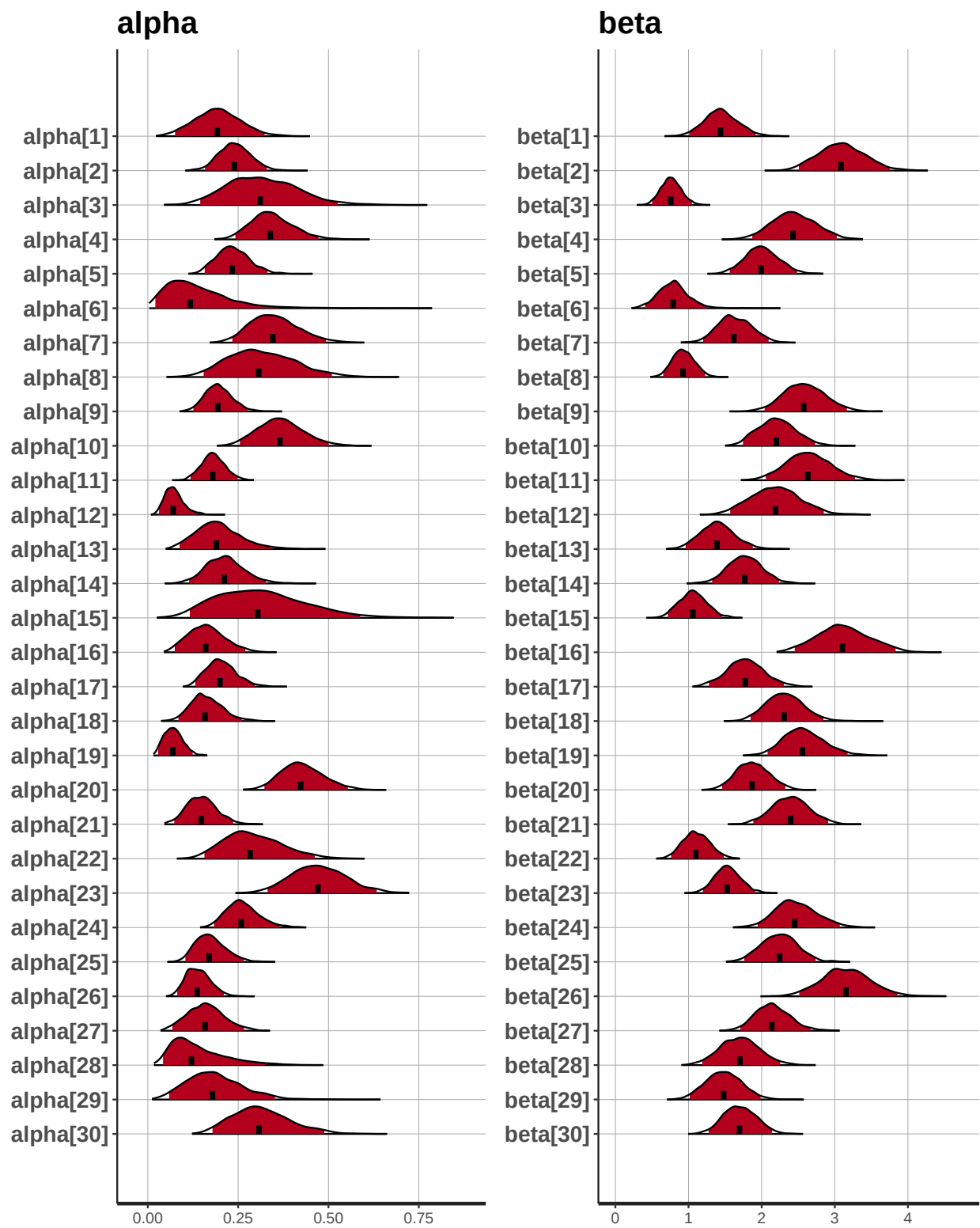




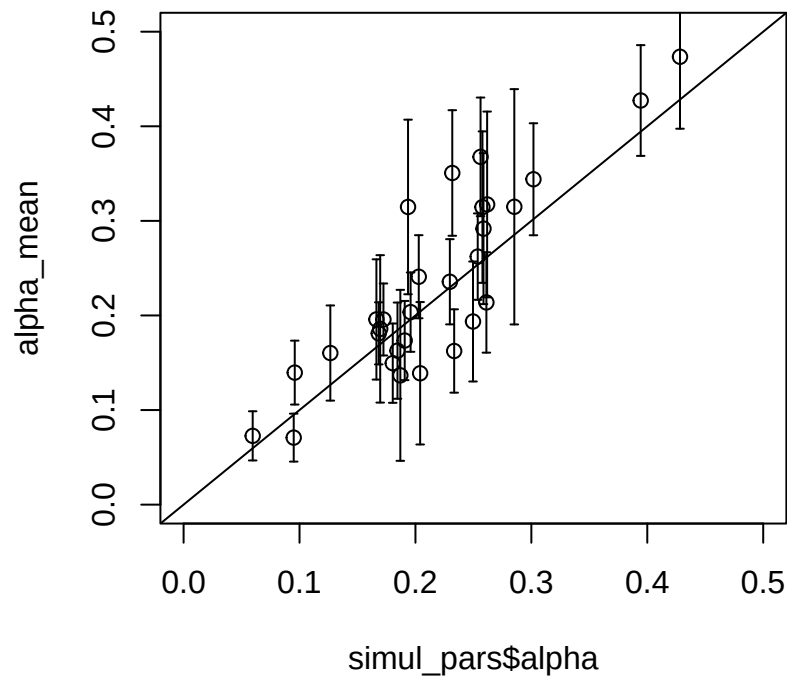
Q2

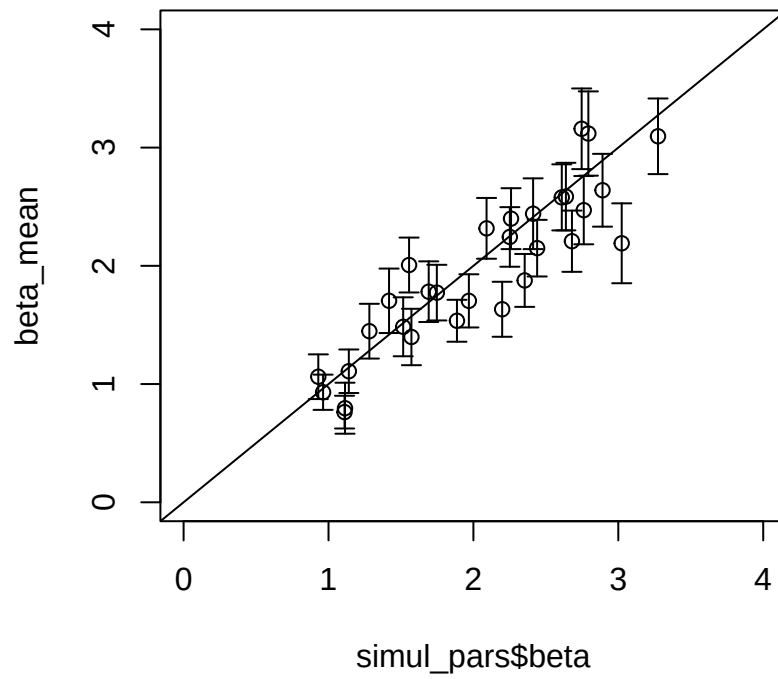
(1) Posterior distributions of the non-hierarchical model

see codes in Q2 folder



(2) Scatterplot of true parameters and estimated parameters





(3) Noticeable difference between Q1 vs. Q2?

The scatterplots of the hierarchical model (Q1) are more concentrated and the results of parameter recovery are better than those of the non-hierarchical model (Q2). This difference is due to the shrinkage

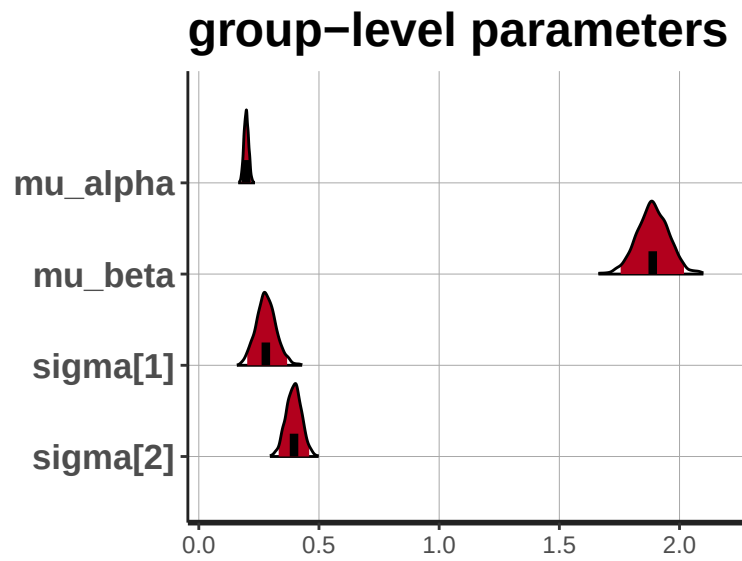
Q3

(1)

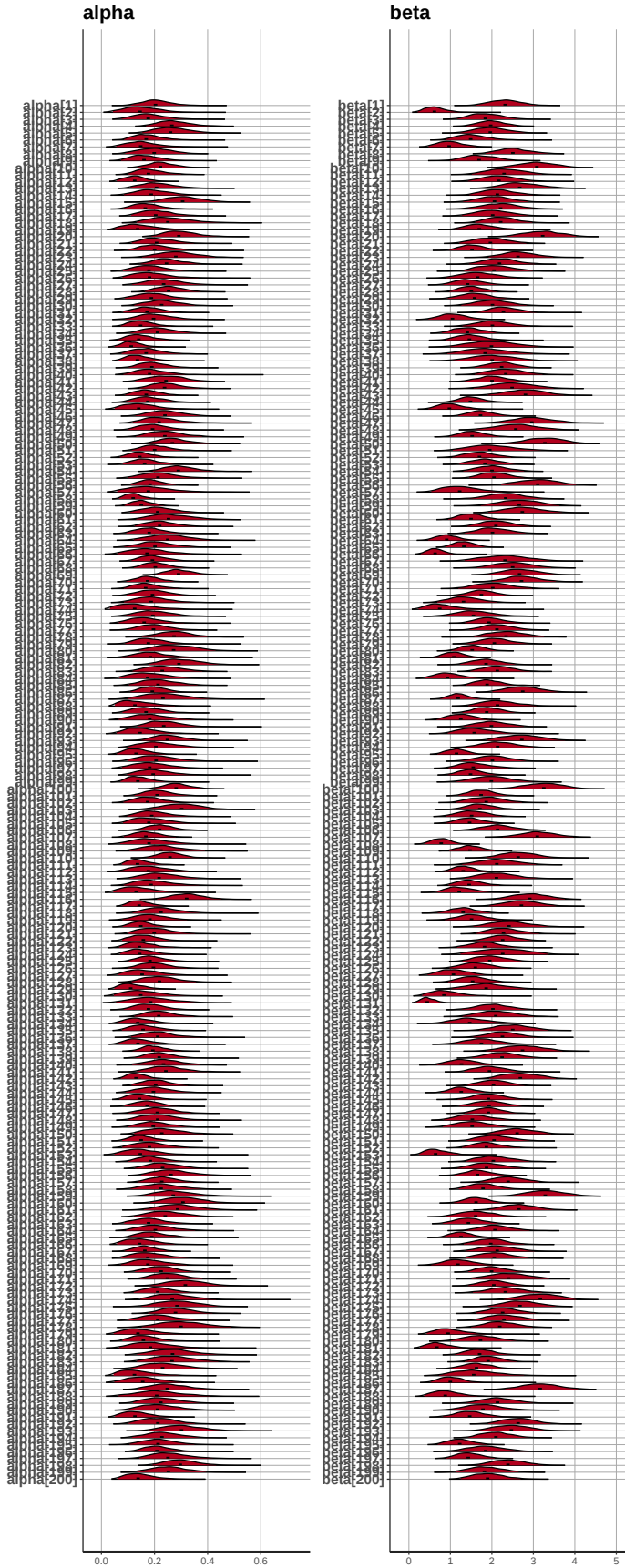
1. For simulation, see `simulate_hw7_model2.R`
2. For parameter estimation, see codes in Q3 folder

(2) Posterior distributions

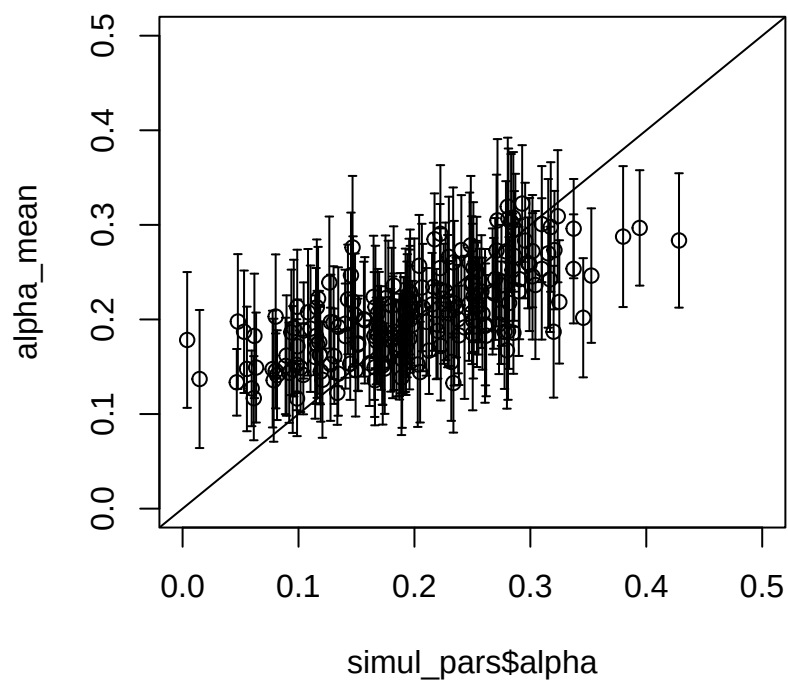
Posterior distributions of group-level parameters

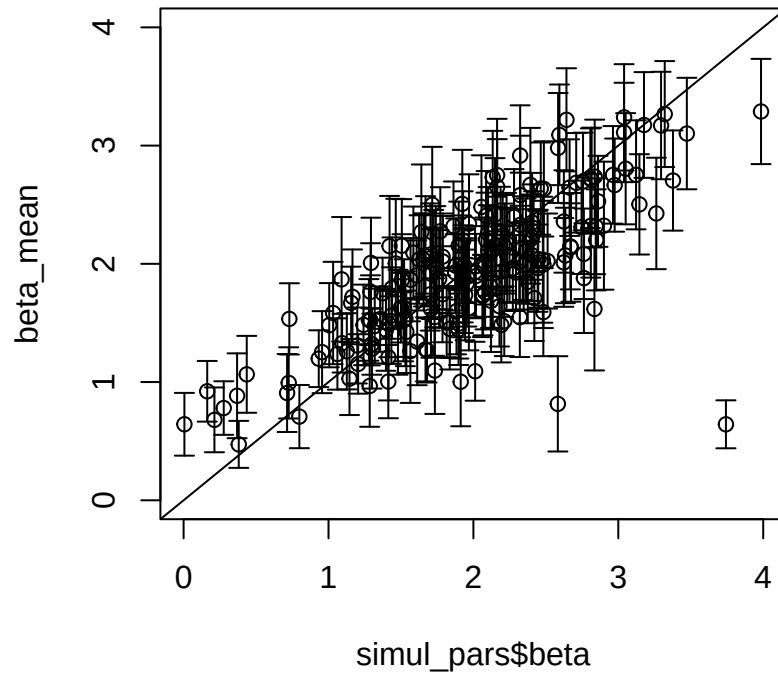


Posterior distributions of individual-level parameters



(3) Scatterplot of true parameters and estimated parameters





(4) Reason for the difference in shrinkage between parameters?

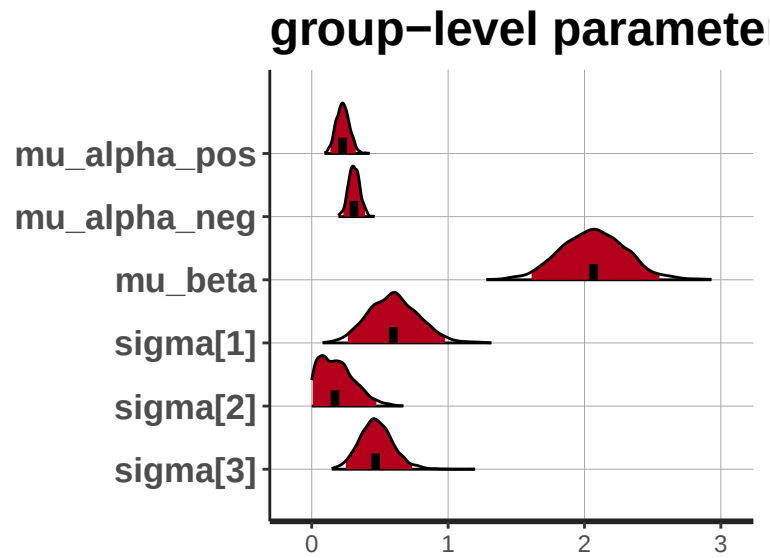
As can be seen in the scatterplots, the estimation of the individual parameter α seems to be more shrunk toward the value of μ_α than parameter β . This is because the degree of shrinkage is greater for small coefficients compared to large coefficients.

Q4

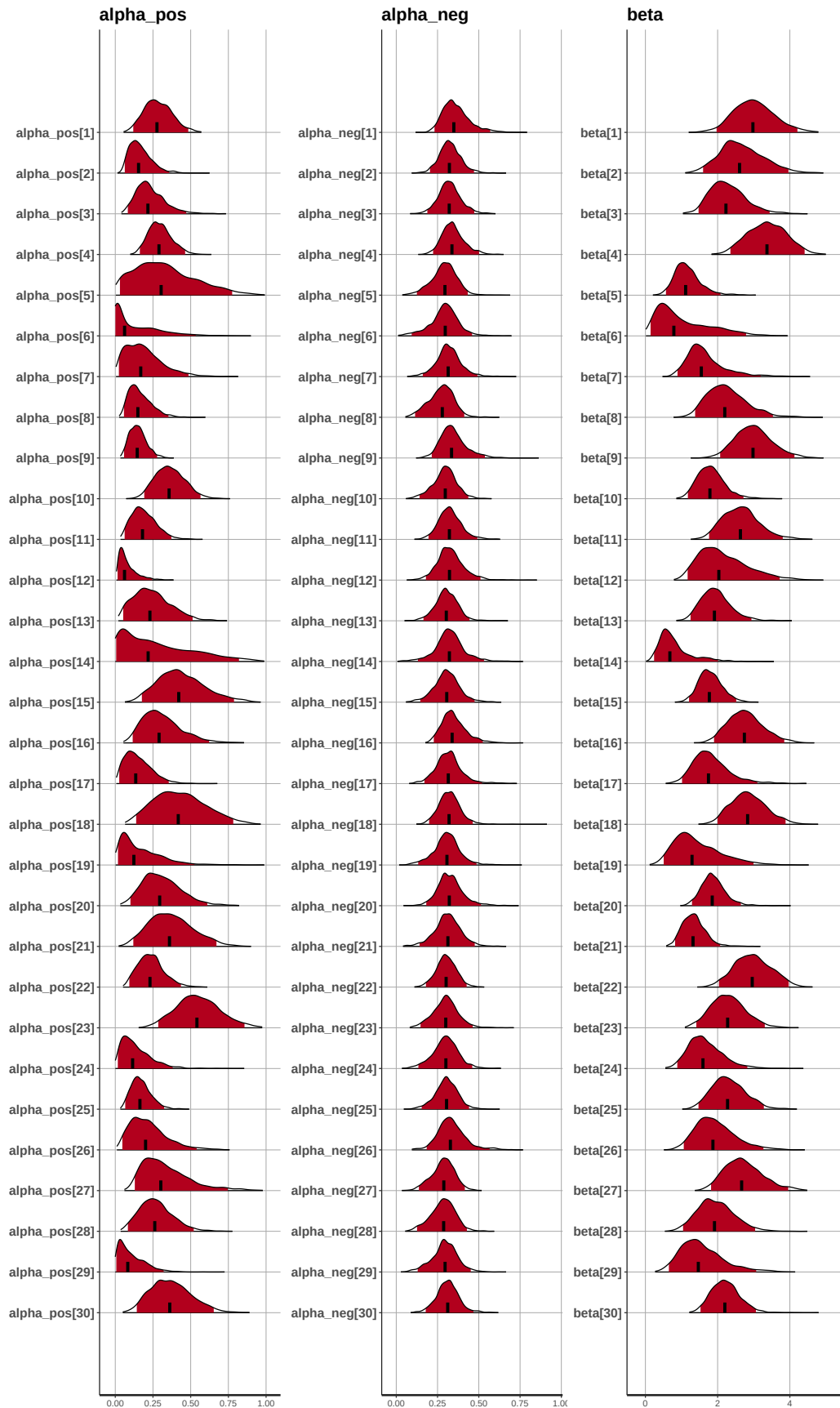
(1)

1. For simulation, `simulate_hw7_model3.R`.
2. For parameter estimation, see `Q4_hw7_model3.R` & `Q4_hw7_model3.stan` in Q4 folder.

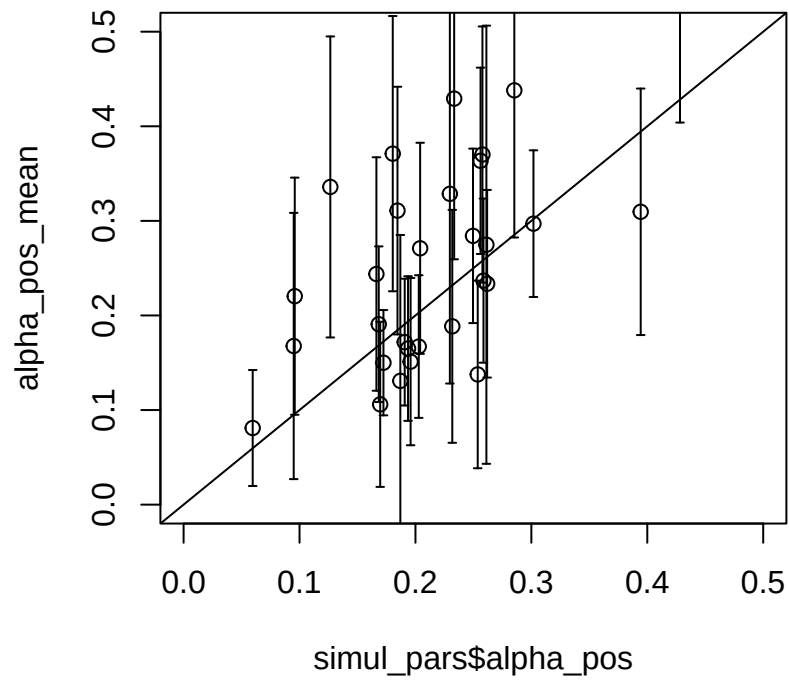
Posterior distributions of group-level parameters

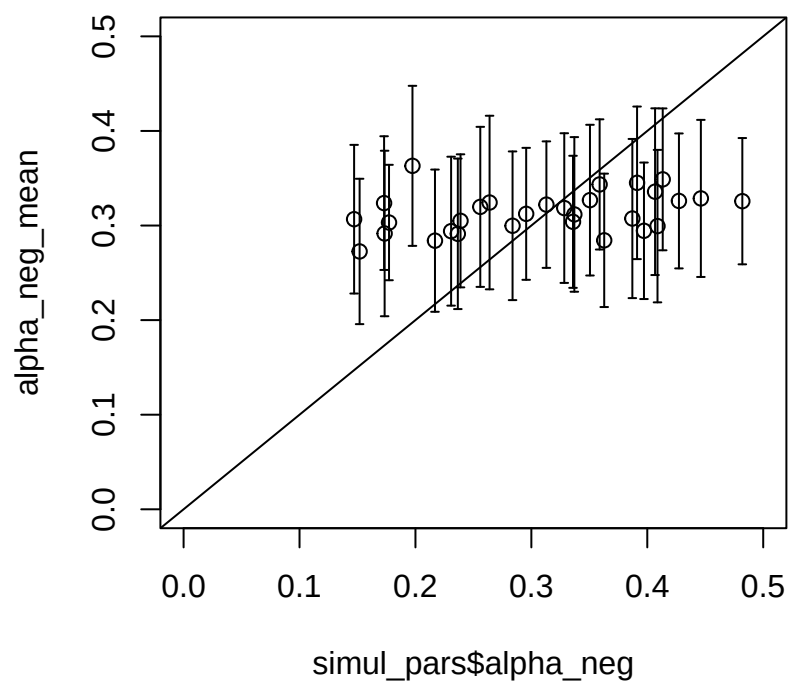


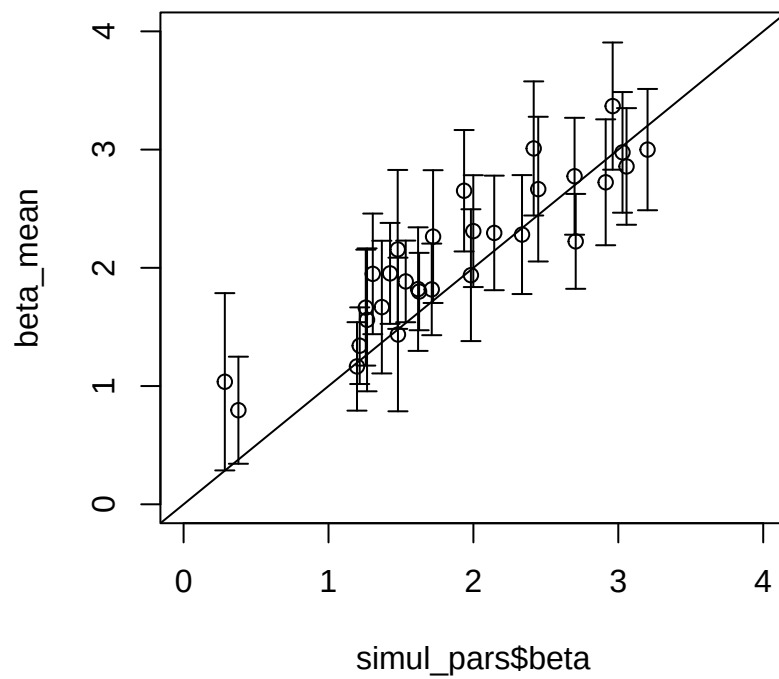
Posterior distributions of individual-level parameters



(2) Scatterplot of true parameters and estimated parameters



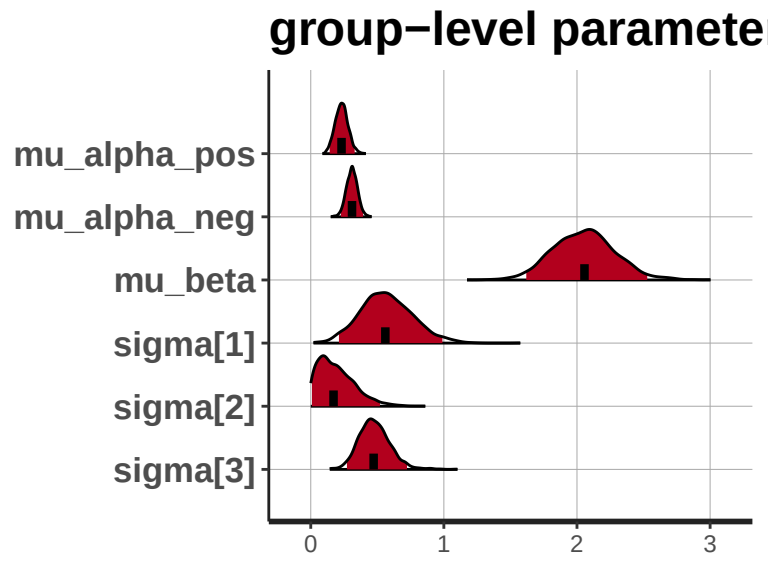




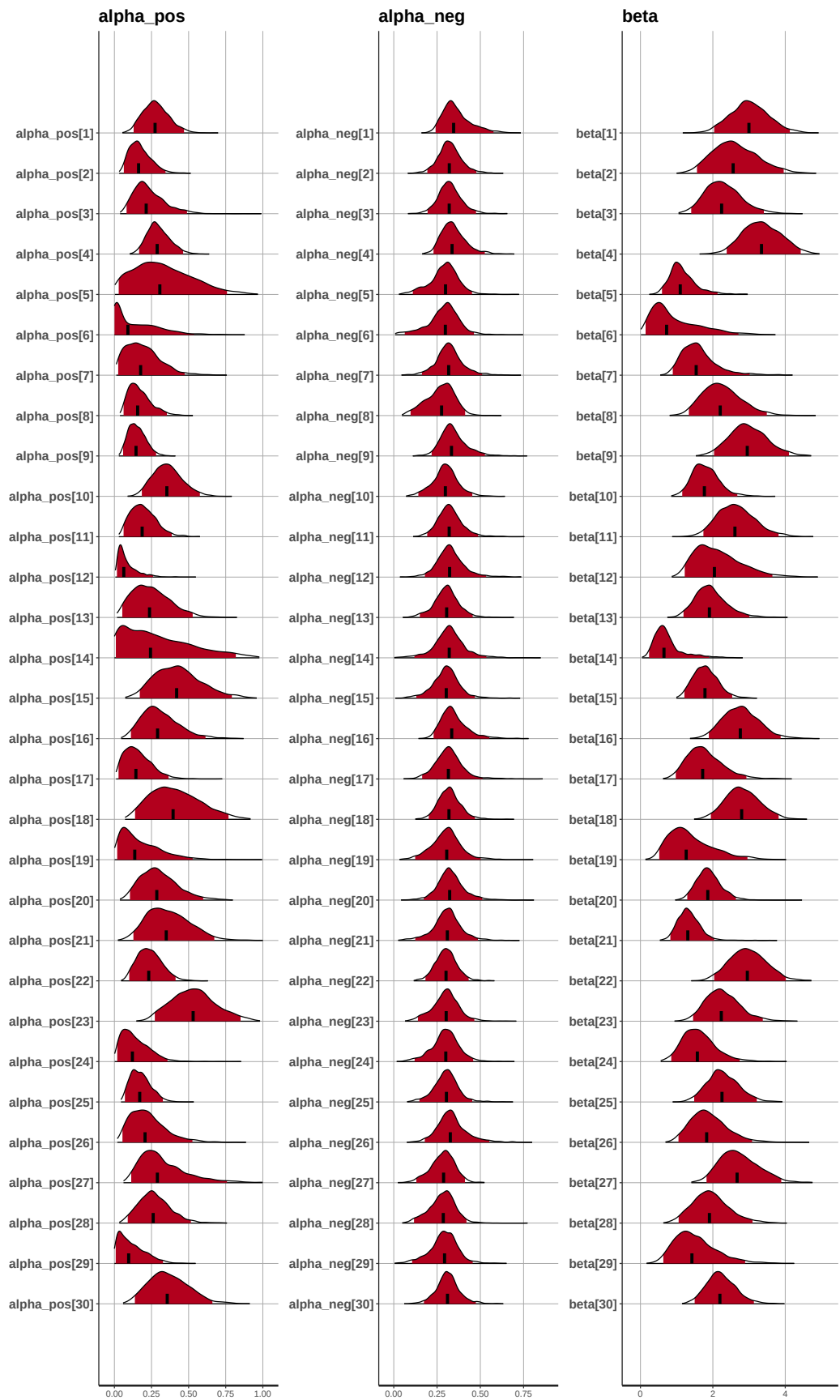
(3)

1. For simulation, see `simulate_hw7_model14.R`.
2. For parameter estimation, see `Q4_hw7_model14.R` & `Q4_hw7_model14.stan` in Q4 folder.

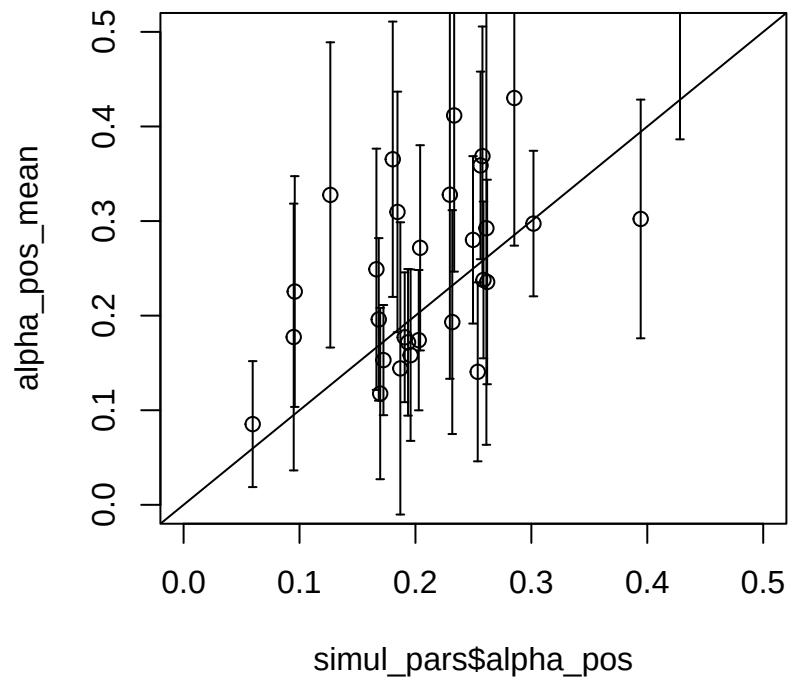
Posterior distributions of group-level parameters

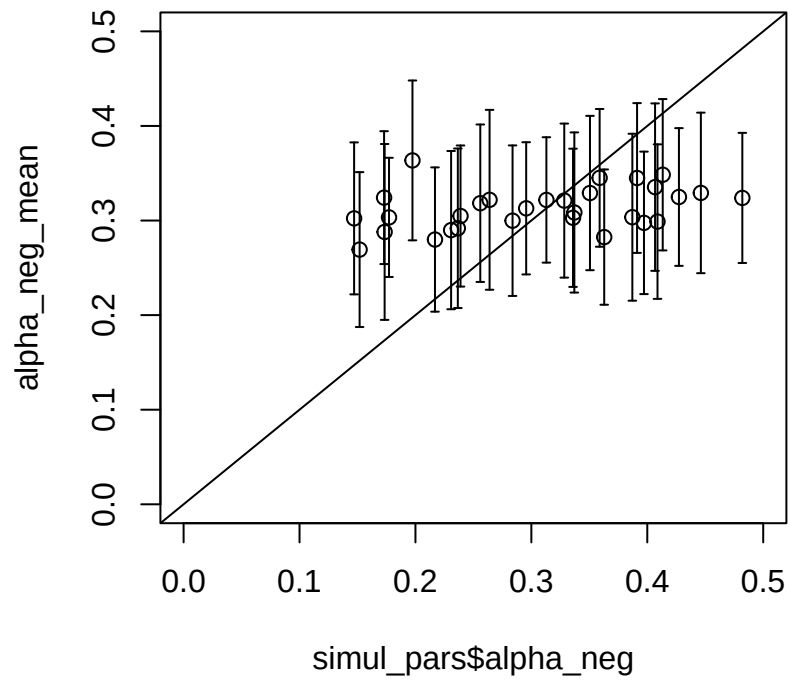


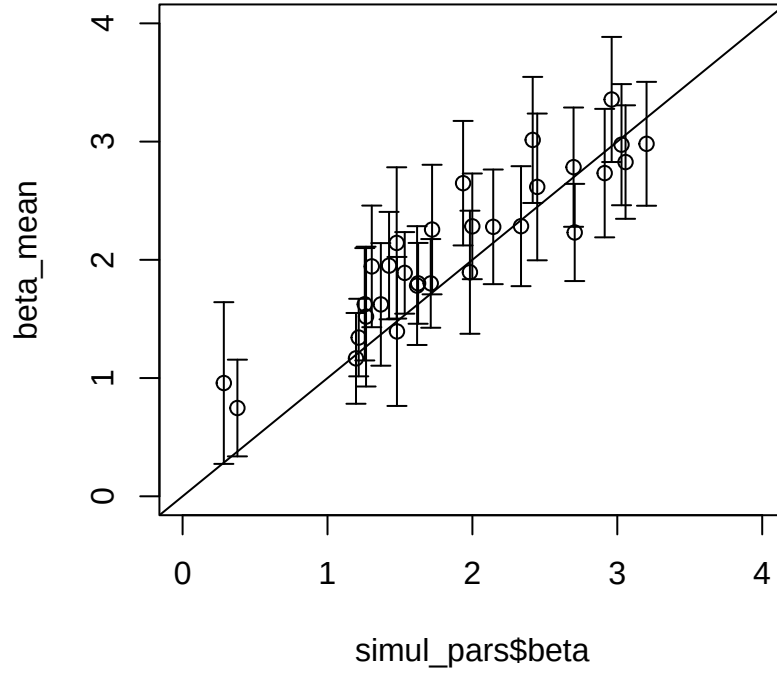
Posterior distributions of individual-level parameters



Scatterplot of true parameters and estimated parameters







(4) Difference between Q4(2) vs. Q4(3)?

Q4(3) showed better parameter recovery results than Q4(2), for the alpha parameters. One reason could be the increase in the number of trials in Q4(3), which increases the information for each individual. The low estimation accuracy of `alpha_neg` is a consequence of setting relatively high `pr_correct` in Q4(2) compared to Q4(3). Thus, there was relatively smaller information for negative PE in Q4(2) while positive and negative PE were fairly well distributed in Q4(3).